Syllabus Details

Syllabus ID:	11585
Syllabus Name:	**Introduction to Computer Science_Nhập môn khoa học máy tính**
Course Name English:	**Introduction to Computer Science**
Subject Code:	**CSI106**
Learning-Teaching Method:	Social Constructivism,Inquiry-Based Learning,In-class lecture, Active Learning
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 5h online + 1h final exam + 99h self-study
Pre-Requisite:	
Description:	This course provides an overview of computer fundamentals. Topics cover all areas of computer science in breadth such as computer organization, network, operating system, data structure , file structure, social and ethical issues. Major Instructional Areas - Introduction to Von Neumann Model and computer components - Numbering system & data representation - Different data types and operations on data - The concepts of computer networking and internet - Operating system - Introduction to basic algorithms and algorithms representation - Introduction to Data structure and File structure - The concepts of database - Software engineering - Computing Security & Ethics - Introduction to Artificial Intelligence (Additional Learning)
StudentTasks:	- Students must attend at least 80% of contact sessions in order to be accepted to the final examination. - Students are responsible to do all exercises, assignments and labs given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Complete Additional Learning (AI For Everyone course in Coursera) and achieve the certificate - Promptly access to the https://flm.fpt.edu.vn for up-to-date course information"
Tools:	
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	765/QĐ-ĐHFPT dated 07/29/2024
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	7/29/2024

4 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Foundations Of Computer Science	Behrouz Forouzan	Cengage	2017	4th	978-1-4737-5104-0	True	True	False	
2	Fundamentals of Information Systems Security		Jones & Bartlett	2016	3rd		False	True	False	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
3	Computer architecture : A quantitative approach	John L. Hennessy, David A Patterson	Morgan Kaufmann	2011	5th		False	True	False	
4	AI For Everyone" course in Coursera https://www.coursera.org/learn/ai-for-everyone/	Andrew Ng	DeepLearning.AI				True	False	True	Additional Learning

13 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand the subsystems of a computer and describe the role of these components
2	CLO2	Know how to convert a number from one base to other base(decimal, binary, octal and hexadecimal)
3	CLO3	Describe how different data type is stored inside a computer as bit patterns and operations on data
4	CLO4	Describe the layers in netwoking and their relationship
5	CLO5	Undestand the role of OS in a computer system
6	CLO6	Define a algorithm and describe tools used to develop algorithms
7	CLO7	Explain how a program in a high-level language is translated into machine language
8	CLO8	Describe the software lifecycle
9	CLO9	Discuss some data structures: Array, Records, Linked list
10	CLO10	Distinguish between text file and binary file and access methods
11	CLO11	Describe and explain operations within a relational database
12	CLO12	Descibe security goals and discuss ethical issues
13	CLO13	Describe artificial intelligence and its basic concepts

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Introduction to the Course 1.Computer Organization 1.1 Von Neumann model	Offline	CLO1	IT	- Chapter 1, text book - Slides: Part 1-1, - EduNext Slot 1 (Q&A)	CSI106	Read & Review Slides Part 1-1; Answer Edunext question in slot 1	
2	1.2 Computer generations	Offline	CLO1	TU	- Chapter 1, text book - Slides: Part 1-2 - EduNext Slot 1 (Q&A)	CSI106	Read & Review Slides Part 1-2 Answer Edunext question in slot 1	
3	1.3 Subsystems and the role of subsystems	Offline	CLO1	TU	- Chapter 1&5, text book - Slides: Part 1-3 - EduNext Slot 1 (Q&A)		Read & Review Slides Part 1-3 Answer Edunext question in slot 1	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
4	1.4 Central Processing Unit	Offline	CLO1	TU	- Chapter 1&5, text book - Slides: Part 1-4 - EduNext Slot 2 (Q&A)		Read & Review Slides Part 1-4 Answer Edunext question in slot 2	
5	1.5 Memory: main memory	Offline	CLO1	TU	- Chapter 5, text book - Slides: Part 1-5 - EduNext Slot 2 (Q&A)		Read & Review Slides Part 1-5 Answer Edunext question in slot 2	
6	1.6 Input/Output subsystems	Offline	CLO1	TU	- Chapter 5, text book - Slides: Part 1-6 - EduNext Slot 2 (Q&A)		Read & Review Slides Part 1-6 Answer Edunext question in slot 2	
7	1.7 Different architectures	Offline	CLO1	TU	- Chapter 5, text book - Slides: Part 1-7 - EduNext Slot 3 (Q&A)		Read & Review Slides Part 1-7 Answer Edunext question in slot 3	
8	2. Numbering systems 2.1 Introduction 2.2 Positional Number Systems	Offline	CLO2	TU	- Chapter 2, text book - Slides: Part 2 - EduNext Slot 3 (Q&A)		Read & Review Slides Part 2 Answer Edunext question in slot 3	
9	2. Numbering systems 2.3 Conversion	Offline	CLO2	TU	- Chapter 2, text book - Slides: Part 2 - EduNext Slot 3 (Q&A)		Read & Review Slides Part 2 Answer Edunext question in slot 3	
10	Lab 1: Guide	Offline	CLO2	TU	TextBook Chapter 2, Instructor manuals			
11	Lab 1: Guide	Offline	CLO2	TU	TextBook Chapter 2, Instructor manuals			
12	Introduction to presentation 1	Offline	One or many of CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7, CLO8, CLO9, CLO10	TU	TextBook Chapter 6→16, Instructor manuals			
13	3.Data storage and Operations on Data 3.1 Storing Numbers	Offline	CLO3	TU	- Chapter 3, text book - Slides: Part 3 - EduNext Slot 5 (Q&A)		Read & Review Slides Part 3 Answer Edunext question in slot 5	
14	3.Data storage and Operations on Data 3.2 Storing Text, Media, Image, Video	Offline	CLO3	TU	- Chapter 3, text book - Slides: Part 3 - EduNext Slot 5 (Q&A)		Read & Review Slides Part 3 Answer Edunext question in slot 5	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
15	3.3 Logic operations	Offline	CLO3	TU	- Chapter 4, text book - Slides: Part 3 - EduNext Slot 5 (Q&A)		Read & Review Slides Part 3 Answer Edunext question in slot 5	
16	3.3 Shift, Arithmetic operations	Offline	CLO3	TU	- Chapter 4, text book - Slides: Part 3 - EduNext Slot 5 (Q&A)		Read & Review Slides Part 3 Answer Edunext question in slot 5	
17	4. Computer Networks and Internet 4.1 Overview 4.1.1 LAN & WAN 4.1.2 TCP/IP protocol	Offline	CLO4	TU	- Chapter 6, text book - Slides: Part 4 - EduNext Slot 6 (Q&A)		Read & Review Slides Part 4 Answer Edunext question in slot 6	
18	4. Computer Networks and Internet 4.2 layers in netwoking	Offline	CLO4	TU	- Chapter 6, text book - Slides: Part 4 - EduNext Slot 6 (Q&A)		Read & Review Slides Part 4 Answer Edunext question in slot 6	
19	Progress test 1	Offline	CLO1, CLO2, CLO3, CLO4, CLO5	TU				
20	5. Operating System 5.1 Introduction 5.2 Evolution	Offline	CLO5	TU	- Chapter 7, text book - Slides: Part 5 - EduNext Slot 7 (Q&A)		Read & Review Slides Part 5 Answer Edunext question in slot 7	
21	5. Operating System 5.3 Components of OS: UI, memory manager	Offline	CLO5	TU	- Chapter 7, text book - Slides: Part 5 - EduNext Slot 7 (Q&A)		Read & Review Slides Part 5 Answer Edunext question in slot 7	
22	5. Operating System 5.3 Components of OS: file manager	Offline	CLO5	TU	- Chapter 7, text book - Slides: Part 5 - EduNext Slot 7 (Q&A)		Read & Review Slides Part 5 Answer Edunext question in slot 7	
23	5. Operating System 5.3 Components of OS: device manager	Offline	CLO5	TU	- Chapter 7, text book - Slides: Part 5 - EduNext Slot 7 (Q&A)		Read & Review Slides Part 5 Answer Edunext question in slot 7	
24	6. Algorithms 6.1 Concepts: input, output, processing 6.2 Three basic constructs	Offline	CLO6	TU	- Chapter 8, text book - Slides: Part 6 - EduNext Slot 8 (Q&A)		Read & Review Slides Part 6 Answer Edunext question in slot 8	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
25	6. Algorithms 6.1 Concepts: input, output, processing 6.2 Three basic constructs	Offline	CLO6	TU	- Chapter 8, text book - Slides: Part 6 - EduNext Slot 9 (Q&A)		Read & Review Slides Part 6 Answer Edunext question in slot 9	
26	6.3 Algorithm representation 6.4 Search Algorithms: linear, binary	Offline	CLO6	TU	- Chapter 8, text book - Slides: Part 6 - EduNext Slot 9 (Q&A)		Read & Review Slides Part 6 Answer Edunext question in slot 9	
27	6.3 Algorithm representation 6.4 Search Algorithms: linear, binary	Offline	CLO6	TU	- Chapter 8, text book - Slides: Part 6 - EduNext Slot 9 (Q&A)		Read & Review Slides Part 6 Answer Edunext question in slot 9	
28	Presentations 1	Offline	Presentation1. One or many of CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7, CLO8, CLO9, CLO10	TU	Slide for presentation			
29	Presentations 1	Offline	Presentation1. One or many of CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7, CLO8, CLO9, CLO10	TU	Slide for presentation			
30	Introduction to Additional Learning: "AI For Everyone" course in Coursera Introduction to presentation 2 (Students have to complete Additional Learning and achieve the certificate before Presentation 2)	Offline	CLO13	TU				
31	7. Programming 7.1 translation	Offline	CLO7	TU	- Chapter 9, text book - Slides: Part 7 - EduNext Slot 11 (Q&A)		Read & Review Slides Part 7 Answer Edunext question in slot 11	
32	7. Programming 7.2 programming paradigms	Offline	CLO7	TU	- Chapter 9, text book - Slides: Part 7 - EduNext Slot 11 (Q&A)		Read & Review Slides Part 7 Answer Edunext question in slot 11	
33	Lab2: Guide	Offline	One or many of CLO6, CLO7	TU	TextBook Chapter 8, 9 Instructor manuals			
34	Lab2: Guide	Offline	One or many of CLO6, CLO7	TU	TextBook Chapter 8, 9 Instructor manuals			
35	8.Software engineering 8.1 The software lifecycle 8.2 Analysis phase	Offline	CLO8	TU	- Chapter 10, text book - Slides: Part 8 - EduNext Slot 12 (Q&A)		Read & Review Slides Part 8 Answer Edunext question in slot 12	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
36	8.Software engineering 8.1 The software lifecycle 8.2 Analysis phase 8.3 Design phase	Offline	CLO8	TU	- Chapter 10, text book - Slides: Part 8 - EduNext Slot 12 (Q&A)		Read & Review Slides Part 8 Answer Edunext question in slot 12	
37	8.4 Implementation phase	Offline	CLO8	TU	- Chapter 10, text book - Slides: Part 8 - EduNext Slot 13 (Q&A)		Read & Review Slides Part 8 Answer Edunext question in slot 13	
38	8.5 Testing phase	Offline	CLO8	TU	- Chapter 10, text book - Slides: Part 8 - EduNext Slot 13 (Q&A)		Read & Review Slides Part 8 Answer Edunext question in slot 13	
39	9. Data structures 9.1 Arrays 9.2 Records	Offline	CLO9	TU	- Chapter 11, text book - Slides: Part 9 - EduNext Slot 14 (Q&A)		Read & Review Slides Part 9 Answer Edunext question in slot 14	
40	9. Data structures 9.3 Linked List	Offline	CLO9	TU	- Chapter 11, text book - Slides: Part 9 - EduNext Slot 14 (Q&A)		Read & Review Slides Part 9 Answer Edunext question in slot 14	
41	9.4 Introduction: Stack, Queue, Tree, graph	Offline	CLO9	TU	- Chapter 11, text book - Slides: Part 9 - EduNext Slot 14 (Q&A)		Read & Review Slides Part 9 Answer Edunext question in slot 14	
42	Progress test 2	Offline	CLO5, CLO6, CLO7, CLO8, CLO9	TU				
43	10.File structure 10.1 Text versus Binary	Offline	CLO10	TU	- Chapter 13, text book - Slides: Part 10 - EduNext Slot 15 (Q&A)		Read & Review Slides Part 10 Answer Edunext question in slot 15	
44	10.File structure 10.2 Access methods	Offline	CLO10	TU	- Chapter 13, text book - Slides: Part 10 - EduNext Slot 15 (Q&A)		Read & Review Slides Part 10 Answer Edunext question in slot 15	
45	11. Database 11.1 Introduction	Offline	CLO11	TU	- Chapter 14, text book - Slides: Part 11 - EduNext Slot 16 (Q&A)		Read & Review Slides Part 11 Answer Edunext question in slot 16	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
46	11. Database 11.2 Database architecture	Offline	CLO11	TU	- Chapter 14, text book - Slides: Part 11 - EduNext Slot 16 (Q&A)		Read & Review Slides Part 11 Answer Edunext question in slot 16	
47	11.3 Database model 11.4 The relationship database model	Offline	CLO11	TU	- Chapter 14, text book - Slides: Part 11 - EduNext Slot 16 (Q&A)		Read & Review Slides Part 11 Answer Edunext question in slot 16	
48	11.5 Database design	Offline	CLO11	TU	- Chapter 14, text book - Slides: Part 11 - EduNext Slot 16 (Q&A)		Read & Review Slides Part 11 Answer Edunext question in slot 16	
49	11.6 Guide do practice set (database)	Offline	CLO11	TU	- Chapter 14, text book - Slides: Part 11 - EduNext Slot 16 (Q&A)			
50	12. Security and Ethical Issues 12.1 Introduction 12.2 Confidentiality	Offline	CLO12	TU	- Chapter 16 & 20, text book - Slides: Part 12 - EduNext Slot 17 (Q&A)		Read & Review Slides Part 12 Answer Edunext question in slot 17	
51	12. Security and Ethical Issues 12.3 Ethical Principles	Offline	CLO12	TU	- Chapter 16 & 20, text book - Slides: Part 12 - EduNext Slot 17 (Q&A)		Read & Review Slides Part 12 Answer Edunext question in slot 17	
52	12.4 Privacy	Offline	CLO12	TU	- Chapter 16 & 20, text book - Slides: Part 12 - EduNext Slot 18 (Q&A)		Read & Review Slides Part 12 Answer Edunext question in slot 18	
53	12.5 Hackers	Offline	CLO12	TU	- Chapter 16 & 20, text book - Slides: Part 12 - EduNext Slot 18 (Q&A)		Read & Review Slides Part 12 Answer Edunext question in slot 18	
54	Presentation 2	Offline	CLO13	TU	Slide for presentation			
55	Presentation 2	Offline	CLO13	TU	Slide for presentation			
56	Review Additional Learning (AI For Everyone course)	Offline	CLO13	TU				
57	Progress test 3	Offline	CLO10, CLO11, CLO12, CLO13	TU			Test yourself	
58	Review	Offline	All CLOs	TU				
59	Review	Offline	All CLOs	TU				
60	Review	Offline	All CLOs	TU				

41 Constructive question(s)

No.	Session No	Name	Details
1	13	CQ5.1	How to store the sign of number on computer
2	13	CQ5.2	Compare and constrast storing text, image, sound and video on computer
3	14	CQ5.3	What is the difference between logic operation at bit level and pattern level?
4	15	CQ5.4	What is the difference between an arithmetic operation and a shift operation?
5	16	CQ6.1	Compare and constrast characteristics LAN and WAN
6	17	CQ6.2	What is the role of TCP/IP protocol suite and their relationship?
7	18	CQ6.3	How to transfer data from one computer to another?
8	25	CQ9.1	Define three constructs: sequence, decision, and repetition. Describe their use in algorithms
9	26	CQ9.2	What is the UML diagram ? How is a UML diagram related to an algorithm?
10	26	CQ9.3	How can we write an example of an algorithm by UML diagram?
11	27	CQ9.4	What is the pseudocode ? How is pseudocode related to an algorithm?
12	27	CQ9.5	How can we write an example of an algorithm by pseudocode?
13	31	CQ11.1	How can machine language, assembly language and a high-level language be distinguished?
14	31	CQ11.2	How can compilation and interpretation be distinguished?
15	32	CQ11.3	How can four basic computer language paradigms be distinguished?
16	32	CQ11.4	What are the similarities and differences between a procedural paradigm with an object-oriented paradigm?
17	35	CQ12.1	What is the software lifecycle in software engineering?
18	35	CQ12.2	What is the development processes? What are the two most common types of development processes
19	36	CQ12.3	What is the purpose of the analysis phase? What are two trends in this phase?
20	36	CQ12.4	What is the purpose of the design phase? What are two trends in this phase?
21	37	CQ13.1	How can the implementaion phase be described? What are the qualitiy issues in this phase?
22	38	CQ13.2	What is the purpose of the testing phase? How can glass-box testing and black-box testing be distinguished?
23	39	CQ13.3	What is an Array as a data structure? How is it used to store a list of data items?
24	39	CQ13.4	What is a Record as a data structure ? How do you think about Record and Array
25	40	CQ14.1	How can we write a program using pointers? What are the similarities and differences among arrays, records and linked lists?
26	41	CQ14.2	What is the concept of an abstract data type (ADT)? What is a Stack?
27	41	CQ14.3	What is a Queue? What are the differences between Stack and Queue?
28	42	CQ14.4	What is a General Tree? What is a Binary Tree? What are the differences and similarities between General Tree and Binary Tree?
29	42	CQ14.5	What is a Binary Search Tree (BST)? What is a Graph? What are the differences and similarities between Tree and Graph?
30	43	CQ15.1	What is the difference between a text file and a binary file?
31	44	CQ15.2	How to retrieve information from the file?
32	45	CQ15.3	How do we operate on a sequential file, an indexed file, or a hashed file?
33	46	CQ16.1	What is a database?
34	47	CQ16.2	Why do we need to use architecture in database system?
35	48	CQ16.3	Identify an advantage that an object-oriented database can have over a relational database.
36	49	CQ17.1	What is the purpose of a database model?
37	50	CQ17.2	How does the relational model provide for data independence?
38	51	CQ17.3	How do you design a good database?
39	52	CQ18.1	What are security goals?

No.	Session No	Name	Details
40	53	CQ18.2	How confidentiality can be achieved using symmetric-key and asymmetric-key cipher?
41	54	CQ18.3	What is the main ideas of aspect security?

4 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Not
1	Group presentation	on-going	2	10.0%	>0	Option 1: 30'/group Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Presentation1. One or many of CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7, CLO8, CLO9, CLO10 Presentation2. CLO13	Option 1: .ppt and oral presentation Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: N/A Option 2 (For Constructivism Approach only): Follow lecturer's proposal	one of the appendices	Each group selects one topic ralated to Addional Learning ((AI For Everyone course in Coursera) to present and has at least two weeks to prepare the presentation Presentation 2: - .ppt and oral presentation - Topics related to Additional Learning (Chapter 18 in Textbook, Deep Learning in Object Detection and Recognition, Speech Emotion Recognition, Traffic Sign Recognition, Federated Learning,...)	The pres is g and stuc com Add Lea For cou Cou ach cert
2	Lab	on-going	2	20.0%	>0	Option 1: 60-90'/each Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Lab1. CLO2 Lab2. One or many of CLO6, CLO7	Option 1: practical exercises Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: N/A Option 2 (For Constructivism Approach only): Follow lecturer's proposal	technical aspect of studied topics	Guided by instructor - Labs are arranged relevant to the chapter theory - Student must finish labs and submit the results to the lecturer in class.	.

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Not
3	Progress Test	on-going	3	30.0%	>0	Option 1: 30'/each Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Test1. CLO1, CLO2, CLO3, CLO4, CLO5 Test2. CLO6, CLO7, CLO8, CLO9 Test3. CLO10, CLO11, CLO12, CLO13	Option 1: MC or suitable format Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: 30 Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Test 1: Chapters 1, 2, 3, 4, 5 , 6, 7 Test 2: Chapters 8, 9, 10, 11, 12 Test 3: Chapters 13, 14, 16, 18, 20	in class, by instructor	Inst sch Prog mus pres the Imp Plar by c the The be t afte lect requ mat Inst resp to re test stud grad
4	Final exam	Final exam	1	40.0%	4	60'	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6, CLO7, CLO8, CLO9, CLO10, CLO11, CLO12	Multiple choices Marked by Computer	60	concepts, algorithms; all studied Chapters .	by exam board, using computer	Stud com Add Lea For cou Cou ach cert orde fina The que be u diffe leas the one